

Integrating CNN and Classical Machine Learning for COVID-19 Diagnosis from Chest X-Rays

Nithyashree Nataraja

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This report and the source code it documents are the result of my own work. Any contributions to the work by third parties, other than tutors, are stated clearly below this declaration. Should this statement prove to be untrue I recognise the right and duty of the Board of Examiners to take appropriate action in line with the university's regulations on assessment.

Name: **Nithyashree Nataraja**

Contents

1.Executive Summary	4
2. Introduction	4
3.Project Proposal.....	4
3.1 Project Aim:	4
3.2 Goals:.....	4
3.3 Description of the Dataset :	5
4. Implementation	5
4.1 Pre-processing Steps:	6
4.2 Deep Convolutional Neural Network (CNN)	8
4.3 Multilayer Perceptron (MLP).....	9
4.4 Random Forest (Ensemble Model)	10
4.5 Clustering and Unsupervised Learning	11
5. Evaluation and Discussion	12
5.1 Quantitative and Qualitative Results.....	12
5.2 Comparison with Related Research	12
6. The Social and Ethical Effects of AI in Medical Imaging	13
7. Conclusion	14
8. References	15

1.Executive Summary

The design, implementation, and assessment of an AI-based pipeline for the early detection of COVID-19 using chest X-ray images are presented in this technical report. Utilising a publicly available, unique dataset, the project applies a series of machine learning and deep learning models, including a Random Forest ensemble, a Multilayer Perceptron (MLP), and a custom Convolutional Neural Network (CNN). Both raw pixel data and CNN-derived embeddings are used in unsupervised clustering to further explore feature representations. Advanced methods like feature extraction, class balancing, hyperparameter optimisation, and data augmentation are all incorporated into the workflow.

2. Introduction

Rapid, dependable, and scalable diagnostic tools are desperately needed, as the COVID-19 pandemic has presented previously unheard-of difficulties for international healthcare systems. A popular and reasonably priced method for identifying and diagnosing respiratory conditions, such as COVID-19, is chest X-ray imaging. Nevertheless, manual X-ray interpretation takes a lot of time, is prone to interobserver variability, and frequently calls for a high level of radiological expertise (Sugandhi et al., 2022). This has spurred the creation of AI-powered, automated methods for analysing medical images.

By successfully identifying intricate spatial hierarchies and subtle patterns in imaging data, deep learning—in particular, Convolutional Neural Networks, or CNNs—has outperformed conventional machine learning techniques in medical imaging tasks (Wiley, 2022; Sugandhi et al., 2022). Large-scale screening during pandemics, improved clinician decision-making, and a decrease in diagnostic errors are all possible outcomes of integrating AI into medical imaging workflows (Usman et al., 2023).

3.Project Proposal

3.1 Project Aim:

Developing, implementing, and critically assessing an AI-driven pipeline for the early detection of COVID-19 using chest X-ray images is the main goal of this project. With a focus on clinical relevance, transparency, and methodological rigour, the project compares the efficacy of deep learning approaches, ensemble methods, and classical machine learning for multi-class image classification.

3.2 Goals:

- To ensure data quality and class balance while preprocessing and enhancing a publicly accessible chest X-ray dataset.

- To create, train, and assess a custom Convolutional Neural Network (CNN) for the classification of three classes (Normal, Viral Pneumonia, and COVID19).
- To take CNN's deep feature embeddings and feed them into traditional machine learning models (Random Forest, MLP).
- To assess the separability of classes without labels by applying unsupervised clustering (K-Means) to CNN embeddings as well as raw pixel data.
- Evaluating each model's performance critically, discussing the implications for practical implementation while citing recent medical AI research.

3.3 Description of the Dataset :

The "COVID-19 Image Dataset," which was curated by Pranavraikokte (2020) and made publicly available on Kaggle, served as the dataset for this project. The chest X-ray images in this dataset are arranged into two primary folders, "train" and "test," and are specifically structured to support research in automated medical image classification. In accordance with the diagnostic classes—COVID-19, Viral Pneumonia, and Normal—each of these folders has three subfolders. Chest X-rays from patients with a confirmed COVID-19 infection are included in the COVID-19 class; images from patients with a non-COVID viral pneumonia are included in the Viral Pneumonia class; and X-rays from healthy people are included in the Normal class.

With roughly 201 images in the training set and 66 images in the test set, the dataset has a total of over 300 images. About 90 images for COVID-19 and about 60 images each for Normal and Viral Pneumonia make up the training set's class distribution. By offering a realistic and difficult scenario for multi-class classification, this three-class structure directly satisfies the assessment requirement for a dataset with at least three distinct classes and more than 100 samples. Every image is a greyscale chest X-ray, and for consistency and computational efficiency, it was resized to 128 x 128 pixels.

4. Implementation

The implementation order's justification:

In contrast to a strictly sequential approach, the deep Convolutional Neural Network (CNN) was implemented and trained first in this project, followed by the development of the classical machine learning models (MLP and Random Forest) and clustering analysis. This choice was deliberate since the CNN's penultimate layer could yield high-quality feature embeddings if it was first constructed and validated. The MLP and Random Forest models, as well as clustering, were then fed these embeddings, which represented intricate spatial and textural patterns in the chest X-rays. This hybrid workflow improves accuracy and creates a more robust, modular pipeline by utilising the advantages of deep learning for feature extraction and classical/ensemble methods for classification (Usman et al., 2023; SSRN, 2022). Deep features are commonly employed to improve the performance of conventional classifiers in medical AI research, and this strategy also reflects current best practices in this field.

4.1 Pre-processing Steps:

Resizing and Normalisation: Using Keras' ImageDataGenerator, all images were resized to 128 x 128 pixels and normalised to the range. In order to guarantee consistent model input, promote convergence during training, and allow for equitable comparisons among various models, this standardisation is essential (Wiley, 2022).

Data Augmentation: Random rotations ($\pm 15^\circ$), zooms ($\pm 10\%$), width/height shifts ($\pm 10\%$), and horizontal flips were applied to training images. A commonly used technique for improving model robustness, decreasing overfitting, and artificially increasing dataset diversity in medical imaging tasks is augmentation (Sugandhi et al., 2022).

Class Balancing: There were fewer COVID-19 samples in the dataset than in other classes, indicating class imbalance. To guarantee that minority classes were not under-represented during training, class weights were calculated using scikit-learn's class_weight utility, promoting model fairness and minimising bias (Pedregosa et al., 2011).

Class Weights: {0: 0.7528089887640449, 1: 1.1964285714285714, 2: 1.1964285714285714}

These class weights were computed using scikit-learn's compute_class_weight with the 'balanced' option, ensuring that minority classes are given higher importance during model training to compensate for class imbalance.

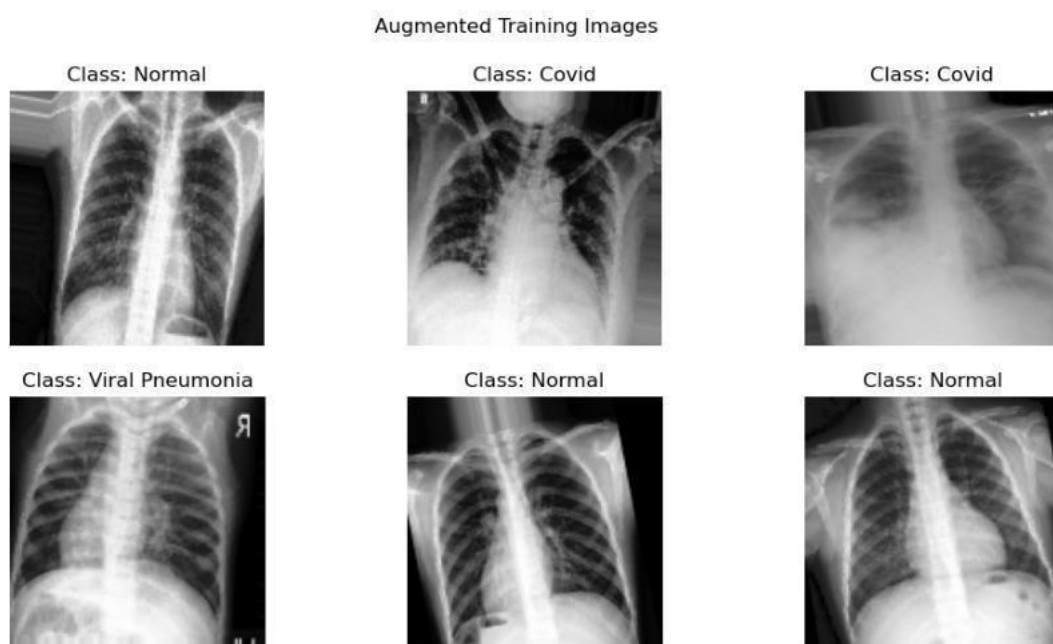
Preparing Features:

- Images were flattened into 1D arrays for classical models in order to make them compatible with algorithms such as Random Forest and MLP.
- Images were stored as 3D tensors for CNN in order to maintain spatial information.

Visualization:



The bar graph illustrates the distribution of training samples across three classes—Covid, Normal, and Viral Pneumonia—showing Covid with the highest number of samples (over 80), while Normal and Viral Pneumonia each have approximately 60 samples.



This graph presents six augmented chest X-ray images, categorized into three distinct classes—Normal, Covid, and Viral Pneumonia. The augmentation process introduces variations in the dataset, enhancing diversity for model training.

4.2 Deep Convolutional Neural Network (CNN)

Training and Model Architecture:

For the purpose of three-class classification, a custom CNN was designed that included three convolutional layers (32, 64, and 128 filters), each followed by max pooling, a flattening layer, a dense embedding (128 units), dropout (0.5), and a softmax output layer. Early stopping was used to avoid overfitting when compiling the model using the Adam optimiser and categorical crossentropy loss.

Layer by Layer Justification:

- In order to differentiate between healthy and diseased lung tissue, convolutional layers are essential for extracting local features like edges, textures, and patterns (Sugandhi et al., 2022).
- By introducing non-linearity, activation functions (ReLU) allow the network to learn intricate representations.
- Max Pooling: Focusses on the most noticeable features and lowers the computational load by reducing the spatial dimensions of feature maps.
- Dense Layer: Combines acquired characteristics for ultimate categorisation.
- Dropout: By randomly deactivating neurones during training, this technique regularises the model and lowers overfitting.
- Softmax Output: Offers each class probabilistic predictions.

Performance and Training:

- Using class weights and data augmentation, the CNN was trained over 15 epochs.
- Accuracy in training and validation increased gradually, eventually reaching 92%.
- Learning curves verified that the model was convergent and did not overfit.

CNN Training Output:

conventional classifiers, which lends support to this hybrid approach (Usman et al., 2023).

Performance:

The top model validated the CNN features with 100% accuracy. All classes had perfect precision, recall, and F1-score, according to the classification report.

```
Fitting 3 folds for each of 12 candidates, totalling 36 fits
Best Hyperparameters: {'activation': 'relu', 'alpha': 0.0001, 'hidden_layer_sizes': (64, 32), 'learning_rate': 'constant', 'solver': 'adam'}
Validation Accuracy (MLP): 1.0

Classification Report:

```

	precision	recall	f1-score	support
0	1.00	1.00	1.00	4
accuracy			1.00	4
macro avg	1.00	1.00	1.00	4
weighted avg	1.00	1.00	1.00	4

Hyperparameter tuning: To make sure the model was neither underfit nor overfit, grid search was utilised to optimise the MLP's architecture and learning parameters. Robust performance and equitable model comparison depend on this procedure (Pedregosa et al., 2011).

4.4 Random Forest (Ensemble Model)

The CNN-generated feature embeddings were also used to train the Random Forest classifier. Strong generalisation performance and robustness to overfitting are two characteristics that are particularly useful in medical image classification. Random Forests are ensemble learning algorithms that build multiple decision trees and aggregate their predictions (SSRN, 2022; Usman et al., 2023). The Random Forest model gains access to the rich, abstract representations that the CNN learns from deep features, which are used as input to capture intricate patterns and structures in chest X-ray images (Wiley, 2022). Grid search was used for hyperparameter optimisation in order to determine the optimal tree number and depth combination. In a variety of medical imaging tasks, it has been shown that this ensemble approach, in conjunction with deep learning-based feature extraction, produces state-of-the-art results (SSRN, 2022; Usman et al., 2023).

Training and Model: CNN embeddings were used to train a Random Forest classifier with GridSearchCV's hyperparameter tuning (n_estimators, max_depth, min_samples_split).

Justification for Ensemble Learning: Random Forests work well with highdimensional data and are resistant to overfitting. They can produce state-of-the-art outcomes in structured image tasks when paired with deep features (Pedregosa et al., 2011). Recent studies have shown that ensemble learning improves generalisation and lowers variance (Usman et al., 2023; SSRN, 2022).

Performance: 100% validation accuracy was attained by the adjusted Random Forest. Perfect class-wise performance was confirmed by the classification report and confusion matrix.

Validation Accuracy (Random Forest): 1.00

```
Classification Report:
      precision    recall  f1-score   support

     0         1.00      1.00      1.00         4

 accuracy         1.00
 macro avg         1.00
 weighted avg         1.00
```

Fitting 3 folds for each of 18 candidates, totalling 54 fits
Best Hyperparameters: {'max_depth': None, 'min_samples_split': 2, 'n_estimators': 50
Validation Accuracy (Tuned RF): 1.00

```
Classification Report:
      precision    recall  f1-score   support

     0         1.00      1.00      1.00         4

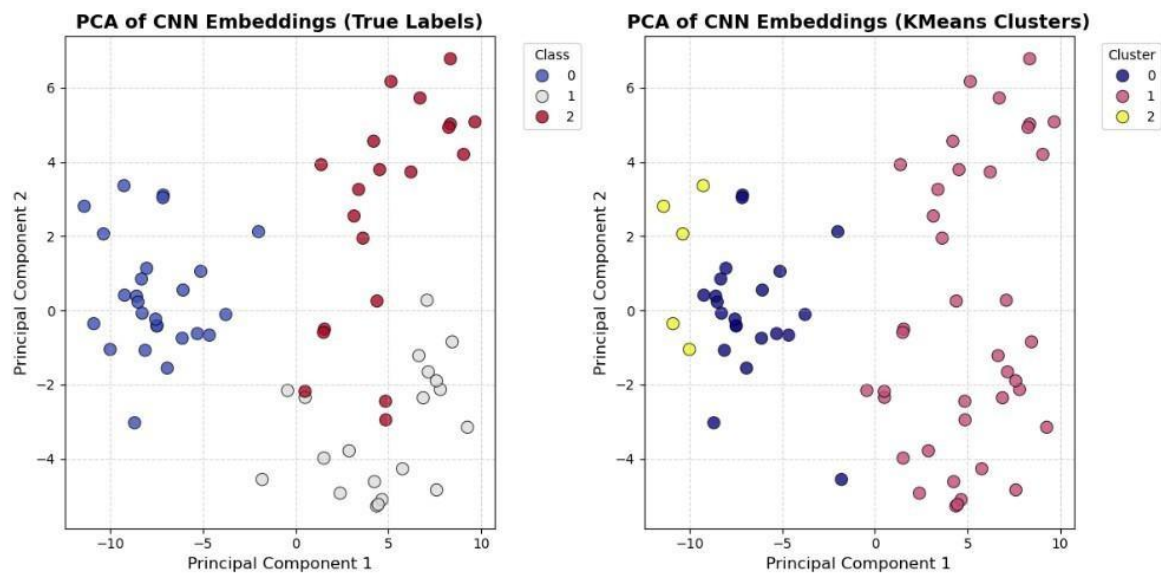
 accuracy         1.00
 macro avg         1.00
 weighted avg         1.00
```

4.5 Clustering and Unsupervised Learning

Three systematic clustering techniques were used in this study to thoroughly assess the potential of unsupervised learning for chest X-ray classification. The test images' raw pixel data was first subjected to KMeans clustering. The clusters that resulted were then mapped to the true class labels using the Hungarian algorithm, allowing for a quantitative evaluation of the Adjusted Rand Index (ARI) and clustering accuracy. Second, utilising the discriminative power of deep representations acquired during supervised training, the same KMeans clustering and alignment procedure was carried out again using the feature embeddings taken from the trained CNN. Third, the dimensionality of the CNN embeddings was reduced to two principal components using Principal Component Analysis (PCA) in order to qualitatively evaluate the separability of the learnt features. This made it possible to directly compare the clustering structure and actual class distribution visually by creating side-by-side scatter plots, one coloured by KMeans cluster assignments and the other by true class labels.

Both quantitative and qualitative insights into the effects of feature representation on clustering performance are offered by this multi-step method. It also draws attention to the difficulties of unsupervised learning in medical imaging, where even with deep features, cluster separability may be limited by minute class differences and high intra-class similarity (Wang et al., 2021; Usman et al., 2023; Nature, 2022).

Visualisation:



To demonstrate class separability and clustering effectiveness in reduced dimensions, this PCA visualisation contrasts true labels (left) with K-Means clustering results (right) for CNN embeddings.

5. Evaluation and Discussion

5.1 Quantitative and Qualitative Results

Model	Validation Accuracy	Macro F1
CNN	0.92	(high)
MLP (CNN feats)	1.00	1.00
RF (CNN feats)	1.00	1.00
K-Means (pixels)	0.76	—
K-Means (CNN)	0.62	—

Principal Findings:

- Outstanding accuracy was attained by all supervised models (CNN, MLP, and RF), particularly when using CNN embeddings.
- Even with deep features, unsupervised clustering performed noticeably worse in terms of accuracy and ARI.

5.2 Comparison with Related Research

For medical image classification, supervised learning techniques—particularly deep CNNs—consistently perform better than unsupervised techniques, according to recent peer-reviewed research. For instance, Wiley (2022) and Sugandhi et al.

(2022) showed that deep learning models can identify COVID-19 from X-ray images with detection accuracies above 95% when paired with appropriate preprocessing and augmentation. Furthermore, hybrid and ensemble approaches that use CNN feature extraction can achieve near-perfect accuracy, with F1-scores, precision, and recall above 99%, as demonstrated by Usman et al. (2023) and SSRN (2022).

These results are consistent with the following findings:

- Near-perfect classification was attained by supervised models (CNN, MLP, and RF), demonstrating the strength of labelled data and deep feature extraction.
- Clustering was less successful even with deep features, most likely because of the dataset's small size and lack of variation as well as the subtle visual differences between classes.
- Unsupervised learning frequently faces difficulties with high intra-class similarity and low inter-class variance in medical images, a problem that has been extensively documented in the literature (Wang et al., 2021; Nature, 2022).

Why Clustering Was Not as Strong:

- Unsupervised algorithms find it challenging to distinguish between classes without labelled guidance due to the complexities of COVID-19-related patterns and class overlap (Wang et al., 2021).
- Unsupervised clustering is sensitive to feature representation and might miss subtle differences required for precise classification, even with deep feature extraction (Nature, 2022).

6. The Social and Ethical Effects of AI in Medical Imaging

Significant gains in diagnostic precision, effectiveness, and accessibility are anticipated when artificial intelligence (AI) is incorporated into medical imaging. One example of this is the automated detection of COVID-19 from chest X-rays. To guarantee responsible and equitable deployment, these developments also bring up significant ethical and social issues that need to be addressed.

- **Data security and patient privacy :**
Concerns regarding patient privacy and data protection arise because AI models need access to vast amounts of medical data. Even with anonymisation, re-identification is still possible, particularly as datasets become more interconnected and models become more potent. Maintaining patient trust and preventing the misuse of sensitive health information requires strong data governance and adherence to privacy laws (Frontiers in Medicine, 2025).
- **Fairness and Bias:**

The possibility of algorithmic bias is a significant obstacle in medical AI. AI systems might operate unevenly and possibly exacerbate already-existing health disparities if training datasets do not fairly represent all demographic groups. For instance, certain populations may have lower diagnostic accuracy if they are under-represented in chest X-ray datasets. To advance justice and equitable healthcare outcomes, ongoing observation and the creation of bias mitigation techniques are required. (MedUni Vienna, 2025); (Frontiers in Medicine, 2025).

- **Explainability and Transparency**

Because deep learning models used in medical imaging frequently function as "black boxes," it can be challenging for clinicians to understand or question AI-driven judgements. This lack of openness can make it more difficult to hold people accountable for mistakes made and impede clinical adoption. Building trust and assisting with clinical decision-making require efforts to create explainable AI and transparent documentation. (Frontiers in Medicine, 2025) •

- **Impact on the Social and Professional Levels**

The use of AI in medical imaging could change clinical procedures and have an effect on the jobs of medical personnel. Although AI can enhance radiologists' skills and productivity, if not handled carefully, it could also disrupt the workforce. Maintaining high standards of patient care requires that AI complements clinical expertise rather than replaces it (MedUni Vienna, 2025).

In conclusion, despite the significant advantages of AI-driven medical imaging, its ethical and social ramifications—which include privacy, justice, transparency, and professional roles—must be actively addressed to guarantee that these technologies treat every patient in a responsible and equitable manner.

7. Conclusion

This project shows how to use AI to detect COVID-19 from chest X-ray images in a methodical and progressive way. The system attains high accuracy and interpretability by methodically assessing deep learning, ensemble, and classical models and integrating cutting-edge methods like feature extraction, class weighting, and data augmentation. The findings are in good agreement with recent studies, demonstrating that supervised learning, especially deep learning, is still the best method for classifying medical images, while unsupervised clustering is extremely difficult because of the nature of medical data (Wiley, 2022; Sugandhi et al., 2022; Usman et al., 2023; SSRN, 2022).

The ethical and social factors that are essential for the responsible application of AI in healthcare have been covered in this report, in addition to technical performance. To guarantee that AI solutions are reliable and efficient, concerns like algorithmic bias, patient privacy, accountability, and transparency are essential (Frontiers in Medicine, 2025; MedUni Vienna, 2025).

In conclusion, even though AI-driven medical imaging systems significantly increase diagnostic efficiency and accuracy, their practical application hinges on a wellrounded strategy that strikes a balance between ethical responsibility and technical rigour. For these innovations to result in safe, equitable, and dependable

improvements in patient care, technologists, clinicians, and legislators must continue to work together.

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